# Find the first step that changes the story

GUIDED NOTES / use with the projected diagram Name _____

## A SAY THE WORD

### Sign / ES: signo / PT: sinal

Direction uses signs; speed is a size.

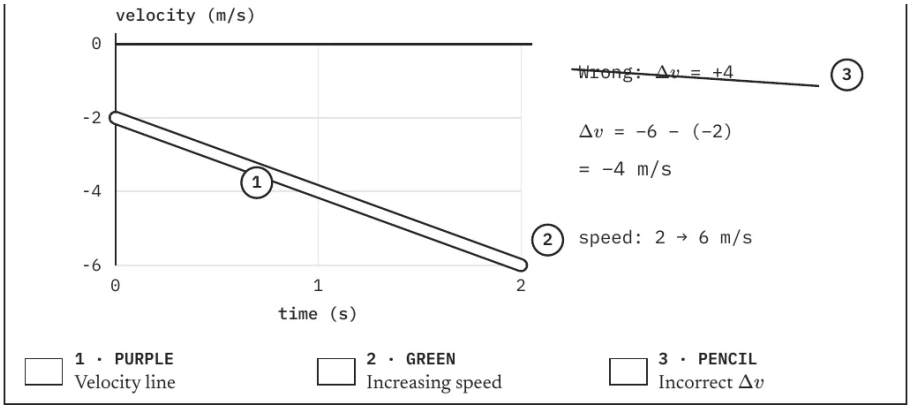
Also on your note page: Quantity, Force, Check.

## B READ THE RELATIONSHIP

### Find the first wrong step, repair it, then recheck

As you read, find: first error · repair · check.

## C TRACE THE ORIGINAL FIGURE




Use the numbered key. Trace one feature. Tell your partner what it means.

## D FOLLOW THE WORKED EXAMPLE

### A 3 kg cart has 9 N right and 3 N left. Someone writes a = 9/3 = 3 m/s² . Explain and repair the first wrong step.

**Given** m = 3 kg right is + forces: +9 N and −3 N

**Rule** Use net force, not the largest force.

**Work** Σ F x = 9 − 3 = +6 N a x = 6/3 = +2 m/s²

**Answer** +2 m/s² Check: ma = 6 N .

## E SAVE FOR THE FINAL MEETING

Before the fictional XJ forecast is shared, check its signs, units, starting coordinates and force sums. Repairing the first wrong step can change every later result.

# Read the physics: Sign

SUPPORTED READING / read one part, then point to the evidence Name _____

## 01 PICTURE IT

Write the positive direction first. Find the first step where the model, sign or starting value went wrong. Changing only the final number can hide the mistake.

## 02 HOW IT WORKS

When forces oppose, give them opposite signs before adding. When reading velocity, compare its size to tell whether speed grows. A more negative velocity can mean a larger speed.

## 03 WHAT TO CHECK

Repair the first wrong step. Then use a sketch, force sum or unit check to test the new reasoning.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

A 3 kg cart has 9 N right and 3 N left. Someone writes $a = 9/3 = 3\ m/s^2$ . Explain and repair the first wrong step. $+2\ m/s^2$ Check: $ma = 6\ N$ .

**Model check: Find the first wrong step, repair it, then recheck.**

## 05 CONNECT TO XJ

Before the fictional XJ forecast is shared, check its signs, units, starting coordinates and force sums. Repairing the first wrong step can change every later result.

SAY IT BEFORE YOU WRITE

What failed first: the rule, the sign or the arithmetic? The first error was ___. I replaced it with ___ because ___.

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Quantity

ES: magnitud física / PT: grandeza física

| Meaning in my words | How can you tell? |
|---|---|
| What do physicists mean by a quantity?<br>In my words, this means ___. | Two labels say 4 m and 4 m/s. What different questions could they answer?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| What label would make "−3" into a useful velocity reading?<br>My example shows ___. | A student writes "force = 6 m." What clue tells you to check their work?<br>I would change ___ because ___. |

## Sign

ES: signo / PT: sinal

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain sign to a friend?<br>In my words, this means ___. | If right is positive, what sign goes on a 3 N force left?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How would you label a leftward velocity of 2 m/s when right is positive?<br>My example shows ___. | Does a negative velocity by itself mean the cart is slowing down?<br>I would change ___ because ___. |

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Force

ES: fuerza / PT: força

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain force to a friend?<br>In my words, this means ___. | When your hand pushes a cart, what pushes and what receives the push?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Which force arrow would you draw while a hand pushes a cart right?<br>My example shows ___. | The hand lets go and the cart keeps rolling. Is the hand still pushing it?<br>I would change ___ because ___. |

## Check

ES: comprobación / PT: verificação

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain check to a friend?<br>In my words, this means ___. | You found displacement using an equation. What feature of a velocity–time graph could check it?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| A cart stays at +3 m/s for 2 s. How can a graph check the result of 6 m?<br>My example shows ___. | Someone copies the same number twice. What could actually catch a wrong answer?<br>I would change ___ because ___. |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Draw a force sum that includes a leftward force. Circle the first place where losing the sign would change your answer.

Labels you can use: first error · repair · check.

First wrong step

Repair

**Tell a partner**

The first error was ___. I replaced it with ___ because ___.

# Stop the faulty graphic

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SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

**Our goal:** Use all forces to calculate acceleration.

**CHOOSE ONE JOB TOGETHER**

**SOFTWARE TESTER** ☐

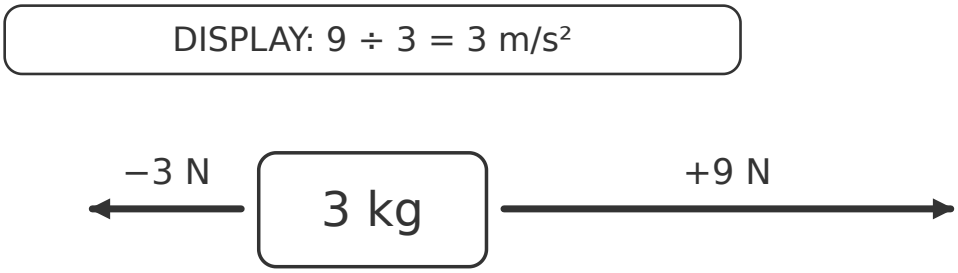
Mark the first wrong calculator step. Send the developer the corrected force sum and expected acceleration.

**FACT-CHECK PRODUCER** ☐

Stop or clear the graphic. Give the anchor a corrected number that viewers can check against the force arrows.

**Word help:** force sum / mass / acceleration

A calculator overlay for a **3 kg** tracker shows **+9 N** and **−3 N**. Its output is "**9 ÷ 3 = 3 m/s²**." It is about to go live.




## 01 OUR FIRST CALL

**Can this calculation go live? Give the corrected result and explain.**

The force sum is ___, so acceleration is ___.

## 02 OUR REVISED CALL

**Wait for the reveal. Keep your first call.**

New clue: _____

Did the developer fix the error? Send a follow-up that names what still needs changing.

Now we would say ___ because ___.

**Switch speakers.** Create a different pair of opposing forces that would expose the same calculator error. Show the correct result.

| ☐ GREEN | ☐ RED | ☐ YELLOW | ☐ BLUE | ☐ PURPLE | ☐ ORANGE | ☐ PENCIL |
|---|---|---|---|---|---|---|
| speeding up | slowing down | steady / balanced | position / displacement | velocity | forces | other marks |